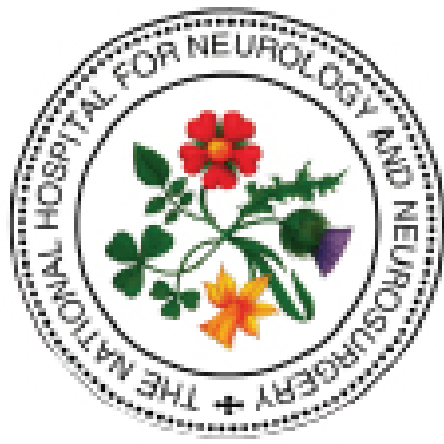




HYDRALAZINE HYDROCHLORIDE for RAPID CONTROL of HYPERTENSION

INDICATIONS FOR USE

Intravenous hydralazine is a potent vasodilator with a rapid onset of action. It can be used as a 2nd line drug to gain rapid control of BP when labetalol has not been effective or is contraindicated.

When administered by IV bolus, hydralazine has an onset of action of 5-20 minutes, peak effect at 10-80 minutes and a duration of action of 2-6 hours. Hydralazine is predominantly metabolised by the liver, with active metabolites excreted by the kidney.

STABILITY AND COMPATIBILITY

- Stable in syringe for 12 hours
- Reconstitute powder with water for injections and then further dilute with 0.9% sodium chloride
- Not compatible with glucose solutions

ACUTE REACTIONS

- Hypersensitivity reactions
- Rapid administration can cause precipitous fall in BP
- Tachycardia, palpitations, angina, flushing
- Headache, dizziness
- Fluid retention
- Gastro-intestinal disturbances

EXTRAVASATION

- Hydralazine has a low pH and may cause venous irritation and tissue damage in cases of extravasation
- If a central venous access device is unavailable, administer via a large peripheral vein monitoring insertion site closely
- Re-site cannula at first signs of inflammation

INTRAVENOUS HYDRALAZINE as 2ND LINE DRUG for RAPID CONTROL of HYPERTENSION

PRESENTATION

- Each 20mg vial contains a powder
- Reconstitute powder with 1mL water for injections to a solution of **20mg in 1mL**
*When a liquid is added to powder during reconstitution the powder **may** create a solution that contains more than the volume added. This extra volume is called the **displacement value** and is specific to each brand of a drug. It is only necessary to consider the displacement value if a part vial is used. For adult patients, displacement is usually insignificant.*
- Requires further dilution with 0.9% sodium chloride before administration
- **NOT** compatible with glucose containing solutions

STEP 1

IV INJECTION

- Dilute the 20mg in 1mL concentrate with sodium chloride 0.9% to a final solution of **20mg in 10mL** (2mg/mL)
- Give slowly over at least **3 minutes**

STEP 2

IV INFUSION

- Further dilute 3 reconstituted 20mg/mL vials with sodium chloride 0.9% to give a final solution of **60mg in 60mL** (1mg/mL)
- Give via infusion pump with a starting rate of **2mg/h** (2mL/h)
- Infusion may be increased by 2mg/h (2mL/h) every 10 minutes titrated to BP target - *allow time for any change to take effect before making adjustments to avoid large swings in BP*
- Maintenance infusion rates usually within the range of 3 - 9mg/h (3-9mL/h) titrated to BP target - *allow time for any change to take effect before making adjustments to avoid large swings in BP*
- The infusion is stable for 12 hours

STEP 3

WEANING

- Decrease infusion by 2mg/h (2mL/h) every 10 minutes
- Oral / NG treatment started, IV therapy weaned and stopped within 2-3 days

RAPID CONTROL OF BLOOD PRESSURE IN ACUTE STROKE

ISCHAEMIC STROKE

Treatment	BP target
IV Thrombolysis	<185/110mmHg
Thrombectomy +/- IV Thrombolysis	<185/110mmHg
Neither IV Thrombolysis nor Thrombectomy	<220/120
NCCU Management	<185mmHg

HAEMORRHAGIC STROKE

<140/90mmHg for at least 7 days
Safe range 130-150mmHg

Aim to achieve **target BP within 1 hour** of starting treatment – *while ensuring it does not drop >60mmHg within 1h*

RAPID BP LOWERING should be avoided if elevated ICP suspected, GCS <6 or neurosurgical evacuation pending

<https://www.ahajournals.org/doi/suppl/10.1161/STR.0000000000000407>

<https://www.strokeguideline.org/app/uploads/2023/04/National-Clinical-Guideline-for-Stroke-2023.pdf>
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